Electronic Supplement to "The Rise of Verbal Reinforcement Learning: A Survey"

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This supplement preserves the extended method catalogs, implementation evidence, benchmark inventories, and secondary comparisons that support the synthesis and decision rules in the main survey.

Additional Key Words and Phrases: verbal reinforcement learning, language feedback, electronic supplement

1 Extended Foundations and Practice

This supplement preserves detailed historical, boundary, taxonomy, practitioner, positioning, and agenda material that supports the synthesis in the main article.

1.1 Terminology and Application Breadth

The phrase *verbal reinforcement* predates machine learning. In mid-twentieth-century behavioral psychology it denoted approving utterances used to shape a subject's responses [79, 130]. In the LLM era, Shinn et al. [222] used "verbal reinforcement learning" for agents that reflect on task feedback and preserve those reflections in episodic memory. The main article generalizes that mechanism-specific coinage through its two membership criteria.

Natural Language Reinforcement Learning [69] is the nearest adjacent formalism. It replaces scalar RL objects with language counterparts, including narrative value functions and language-space policy iteration. Earlier work on language-informed RL instead treated language as a task input or auxiliary knowledge source [20, 166]. The historical shift is therefore from language informing a scalar-reward learner to language serving as evaluation, revision, memory, and sometimes the optimizer trace itself.

The application surface is broad. Language grounds high-level instructions in physical affordances [3]; self-reflection and iterative critique improve program generation [175, 222]; and verbal-feedback mechanisms now appear in coding agents [118, 301], scientific discovery [19, 163], robotics [251], mathematical reasoning [91, 156, 218], clinical decision support [110, 362], and education [4, 201]. These examples establish breadth, not uniform maturity or evidence quality.

1.2 Neighboring Formalisms and Extended Boundary Cases

Two neighboring formalisms clarify what the framework adds. A feedback-conditional policy [168] learns a response distribution conditioned on feedback and therefore keeps the feedback string inside the objective. Memento [354] instead makes memory the learned object in a memory-augmented Markov decision process, with adaptation occurring through case writing and retrieval rather than fine-tuning. CoALA [235] organizes the location of memory, action, and decision modules inside an agent. The signal-centric view is complementary: it classifies what flows through those modules and what the signal modifies.

The POMDP notation has hard limits. Open-ended web and tool-use tasks may provide neither an explicit transition kernel nor an executable reward. Dialogue may lack a compact Markov state, and persistent memory changes later episodes, weakening stationarity. Multi-turn language interaction also has two levels, with utterance decisions layered over token decisions [1, 364]. In these regimes, the formalism supplies vocabulary rather than permission to import POMDP guarantees.

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Table 1. Detailed overview of the three verbal reinforcement learning pillars.

Axis	Grounding signal	Deliberative feedback	Learning signal
When	Problem-definition time	Inference time	Training time
Modifies	Task, state, action, and reward specification	Outputs, context, or memory	Model weights and training data
Persistence	Defines the task	One episode or persistent memory	Future interactions
Mechanisms	Environment and reward specification	Prompting, critique, memory, search	Conditioned or filtered training; on-policy and multi-agent RL
Bottleneck	Grounding fidelity	Feedback quality and compute	Signal compression and filter quality

Membership is adjudicated by tracing dataflow to the scalarization boundary. A rationale-annotated preference qualifies when the rationale enters a revision or training objective, but not when only its preference bit survives. A step-level critique qualifies when downstream machinery converts its content into credit [288]; it reduces to scalar process supervision when only per-step scores survive [147]. A generative process reward model may reason in language before emitting a score [348]. Its linguistic structure is consumed inside reward computation, while the optimizer still receives scalars. Routing the critique itself into revision or supervision moves the boundary later [267].

Current theory covers only selected regimes. The transfer-eluder analysis of Xu et al. [296] separates rich feedback from scalar reward under assumptions that make latent-objective learning possible. The hierarchical analysis of He et al. [92] shows that language-mediated planning still requires exploration. Neither result proves that arbitrary critique, memory, or language-shaped reward improves a policy. No measurable certificate yet reveals how much of a linguistic artifact a consumer actually exploits, and no general bound covers self-generated, noisy, strategic, or adversarial feedback.

1.3 Detailed Taxonomy Aids

Table 1 expands the compact three-pillar definition by making persistence, mechanisms, and bottlenecks explicit.

Temporal scope is preferable to source or modality because the same sentence can produce different effects at different points of use. "The gripper pressure is too high" constrains the problem when embedded in a task specification, repairs a plan when delivered during execution, and shapes learned behavior when consumed by a reward-code generator. A source taxonomy would place all three together even though their failure modes differ. Temporal scope instead groups grounding fidelity, deliberative circularity, and training-time information loss with the mechanisms that must address them.

1.3.1 Working Example: A Coding Agent. Consider an agent resolving a GitHub issue [264]. The issue description supplies the goal, repository files form the observable state, and tests define success. During execution, tracebacks and test messages act as deliberative feedback: the agent reads them and revises its patch. If those trajectories later enter fine-tuning, the same messages become learning signals. This is coexistence, not a required pipeline. A traceback is deliberative

Table 2. Practitioner guidance: when to use language in each role, what it costs, and what to monitor.

Language as	Preconditions	Latency and cost	Dominant risk	Representative
Reward	Success expressible as checks; simulator or executor available; training budget accessible	Offline compilation; cheap per episode	Objective misspecification; reward hacking	Eureka [173]; Text2Reward [289]
State or task grounding	Raw observations complex or task underspecified; language abstracts them faithfully	Per-step description cost	Information loss; state aliasing	ALFWorld [223]; SayCan [3]
Action guidance	Open action space; actions executable and checkable	Per-step generation cost	Exploration collapse; invalid actions	ReAct [309]; ProgPrompt [226]; CodeAct [262]
Deliberative feedback	No weight access; iteration affordable; grounded critic available	Multiplies inference cost	Circular critique; unbounded loops	Self-Refine [175]; CRITIC [77]; ToT [308]
Memory	Tasks recur; lessons reusable	Cheap writes; retrieval overhead	Error persistence; poisoning	Reflexion [222]; ExpeL [347]; Voyager [251]
Training signal	Persistent improvement needed; feedback plentiful; weights accessible	Highest upfront cost	Sycophancy; filter bias	FCP [168]; STaR [321]; generative PRMs [348]

when the running agent conditions on it and a learning signal when a training process consumes it. Classification follows each use of the signal, not the identity of the surrounding system.

1.4 Extended Practitioner Matrix

Table 2 expands the decision tree with preconditions, cost, and monitoring obligations. Mechanism selection precedes optimizer selection because each feedback form restricts the compatible update families.

1.5 Detailed Positioning Against Prior Surveys

Two works anchor the history. Luketina et al. [166] treated language as information supplied to a scalar-reward learner. Natural Language Reinforcement Learning [69] extends RL components into language space but remains one value-centric formal instantiation. The signal-centric framework covers both while distinguishing the stage and consumer of each artifact.

A mechanism-centered cluster fixes the feedback type and organizes its use. Pan et al. [193] classify correction timing, Kamoi et al. [120] analyze when self-correction succeeds, and Zhang et al. [339] treat feedback within test-time scaling. Fernandes et al. [71] provide the closest prior signal typing through feedback format, objective, and use, but predate agentic memory, rubric rewards, and verifiable-reward training.

A component-centered cluster organizes memory, self-evolution, or optimization location. Memory surveys treat persistence as a module [343]; self-evolution surveys classify the component that changes [66, 75]; and agent optimization surveys split parameter-driven from parameter-free updates [59]. The signal-centric view instead follows one critique across context revision, memory, and weight updates.

A reward-centered cluster begins after scalarization. Surveys of RL-enhanced language models and preference optimization emphasize RLHF, RLAIF, or DPO [258, 287]. Other treatments organize alignment by reward design [115], learning by reward model and stage [278], rollout design by pipeline location [238], or credit assignment by granularity and method [328]. These views provide depth after a reward or training mechanism has been chosen. The main article's decision rule operates earlier by asking whether language should remain language and which consumer should receive it.

1.6 Supporting Evidence for the Research Agenda

(1) **Sample efficiency.** The transfer-eluder result separates rich feedback from reward [296], but intermediate formats such as rationale-bearing preferences and step critiques lack comparable complexity measures. Noisy-oracle analyses could connect critic error rates to learning efficiency [151, 232].
(2) **Optimality under language-shaped rewards.** Compiled and rubric rewards place specification error inside the optimized objective. Existing reward-hacking evidence [24] does not yet yield conditions under which the intended behavior remains optimal.
(3) **Temporal credit assignment.** Scalar process rewards attach values to steps [147]; textual gradients attach critique to spans [252]. A general rule connecting free-form critique to responsible decisions remains absent.
(4) **Information retention.** A rate-distortion account could measure what a scalar score or preference label discards relative to its generating critique, clarifying when DPO [203] suffices and when feedback-conditioned training [168] retains necessary structure.
(5) **Provenance and robustness.** Adaptive attacks bypass feedback defenses [325], including injected tool output [220] and poisoned memory [54]. Systems need source authentication, trust assignment, and benchmarks that score adversarial feedback directly.
(6) **Quality versus utilization.** Models may fail to use accurate feedback [117]. Producer-side benchmarks exist [134, 151, 350], but utilization still lacks a standard protocol.
(7) **Feedback-tuned models.** Dedicated critics can outperform generator self-critique [179, 260]. Training objectives should target localization, actionability, and calibration rather than only verdict accuracy.
(8) **Machine-readable interfaces.** Ambiguous tool output causes agent failures [11]; agent-oriented interfaces help [301]; and tool quality bounds refinement [186]. Interfaces should separate task errors from infrastructure failures.
(9) **Inference versus training.** The transient-versus-durable distinction for search rewards [270] should extend to every signal. Persistent or replayed memory [343] makes the boundary particularly difficult.
(10) **Transfer and affected parties.** Self-correction depends on reliable external signal or favorable tasks [120], while incorrect feedback induces sycophancy [216]. Embodied and high-stakes deployment therefore requires transfer studies and explicit analysis of who bears failures.

1.7 Extended Limitations and Deployment Evidence

Language feedback has failure modes that are not yet bounded. A specification compiled from language can be executable yet encode the wrong objective, and policies exploit omissions in rubrics and verifiers [176]. Generative verifiers can be fooled by superficial "master keys" [349], while preference models may reward persuasive agreement over correctness [216].

Grounding remains weak in sequential environments. Vision-language judges may recognize objects while missing task order [280], and single-frame success detectors miss failures expressed through motion [114]. Text2Reward, DrEureka, and RACER demonstrate real-robot deployments [50, 174, 289], but these platform-specific results do not establish broad transfer. High-stakes harms are also no longer hypothetical: sycophantic answer changes appear in medical-advice tasks [68], and injection attacks affect tool-calling, coding, and computer-use agents [62]. These results motivate stronger claims only after systematic cross-platform evaluation.

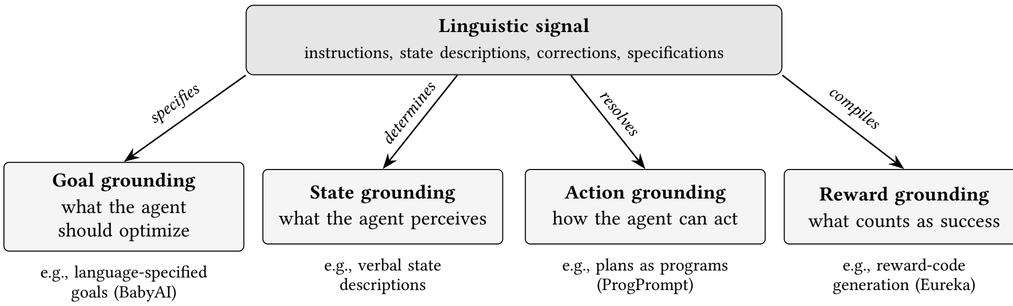

Fig. 1. Grounding signals map verbal input onto four decision-problem components at specification time.

2 Extended Pillar Catalogs

This supplement preserves the extended system catalogs, secondary variants, and benchmark evidence supporting the four pillar chapters in the main article. The organization follows those chapters, while every label is local to the supplement.

2.1 Extended Catalog: Language as Grounding Signal

2.1.1 Historical lineage and goal grounding. Grounding language in sequential decision making predates large language models. Mapping instructions to actions was formulated as a reinforcement-learning problem before pretrained interpreters existed [20, 181], policy sketches used linguistic annotations to decompose multitask learning [6], and agents acquired grounded word meanings in simulated three-dimensional worlds [94]. The earlier literature separated language-conditional from language-assisted reinforcement learning [166], while language-conditioned imitation extended grounding to robot control [169]. Current systems inherit much of the interpreter from pretraining and use it to compile specifications.

Goal grounding requires parsing instructions into objects, relations, and target conditions [27], filtering candidate plans by executability [3, 56], and decomposing goals into verifiable subgoals. Secondary variants include multi-agent coordination [342], hierarchical goal decomposition through offline reinforcement learning [102], and policies trained directly from language specifications [140]. BabyAI isolates compositional generalization in gridworlds [42]; current models still show incomplete coverage under compositional constraints [209], while compositional policy representations improve sample efficiency on unseen combinations [46].

2.1.2 State and action grounding. Verbal state descriptions are native to text environments [47, 307], but they also arise whenever visual or physical state is summarized into language. ALFWorld shows transfer from verbal descriptions to embodied three-dimensional settings [223]. Later work adds closed-loop verbal state feedback for robot planning [18] and scene-graph-structured context [106]. These systems expose a recurring limitation: verbalization can silently violate the Markov property when omitted details carry decision-relevant history.

Action grounding ranges from corrective planning to compilation into programs. Inner Monologue feeds verbalized success detection and scene description back into planning [109], while concise corrective utterances can amend a local error without discarding the rest of a plan [217]. ProgPrompt represents available actions and objects in program-like form [226]; VoxPoser converts language into three-dimensional value maps consumed by a motion planner [108]. Secondary systems combine visual cues with grounded replanning [126], detect failures through constraint-aware visual programs [352], span physical, logical, and semantic corrections across control levels [119], or join language reasoning with feasibility checking in task-and-motion planning [340]. RT-H inserts fine-grained language motions between task and actuation so human corrections arrive in the policy's own intermediate action space [15].

2.1.3 Reward-grounding systems and evidence. Language-to-reward methods differ in how directly text specifies the executable objective. Eureka evolves generated reward programs and uses training statistics as textual reflection [173]; Text2Reward generates dense, human-refinable code [289]. DrEureka extends generation to domain-randomization settings needed for simulation-to-real transfer [174]; CARD refines reward code from trajectory feedback without retraining a policy at every iteration [236]; PROF adapts the mechanism to offline settings [237]; and reward functions can be generated from video demonstrations [322].

Structured alternatives let language select reward parameters rather than write unrestricted code. Language to Rewards maps instructions and corrections to parameters consumed by a real-time optimizer and outperforms a primitive-skill baseline based on Code as Policies [145, 315]. When no compilable state is exposed, a learned judge grounds the objective in observations. RL-VLM-F uses vision-language preferences over image pairs [266], while purpose-trained visual reward models aim for broader reward provision [139, 213].

Reward-grounding errors are specification failures rather than ordinary estimation noise. A syntactically valid reward can encode the wrong objective and then be optimized exactly. Iterative reflection searches reward-function space, and related methods use language to redistribute sparse rewards. LaRe constructs symbolic latent-reward evaluators from language [202]; Agent-RLVR instead densifies a sparse verifier with teacher-style guidance on software-engineering tasks [49]. Controlled evidence also shows that diverse and informative feedback improves transfer and adaptation [281]. These results motivate explicit checks for executability, compositionality, and fidelity before downstream policy performance is interpreted.

2.2 Extended Catalog: Language as Deliberative Feedback

2.2.1 Self-critique and external grounding. Self-Refine formalized iterative self-critique and revision [175], building on Constitutional AI's demonstration that written principles can guide revisions [10]. A separately trained critic improves reasoning relative to a shared generator-critic loop [199]. Other evaluations show that self-correction degrades when the critique source shares the generator's failure modes [104, 120], especially for smaller models [187]. Socratic decomposition creates independently checkable steps [218], and parallel refinement widens the candidate pool [257], but neither supplies knowledge absent from the model.

External grounding anchors critique in deterministic evidence. Toolformer established the broader tool-use setting [212]; execution traces, tests, search results, and API responses then became reusable feedback sources [35, 78]. CRITIC, SWE-agent, and executable-code agents use such evidence to produce corrected outputs [77, 262, 301]. Reinforcement learning can train a model to exploit the same feedback more effectively [76]. The system cost shifts from critic circularity to infrastructure, latency, and correct interpretation of the external trace.

2.2.2 Debate systems and oversight evidence. Multi-agent debate assigns parallel model instances distinct perspectives or roles [98]. Identical models can improve through critique exchange [60], role diversity can strengthen evaluation [26], and models can be trained to revise under conflicting opinions [153]. Adaptive heterogeneous groups also improve mathematical reasoning [360].

The economic evidence is mixed. Debate provides limited gains over single-agent scaling on mathematical reasoning but becomes more useful as task difficulty rises and base capability falls; on safety tasks, collaborative refinement can increase vulnerability while diverse groups reduce attack success [306]. A broader evaluation across debate methods, benchmarks, and foundation

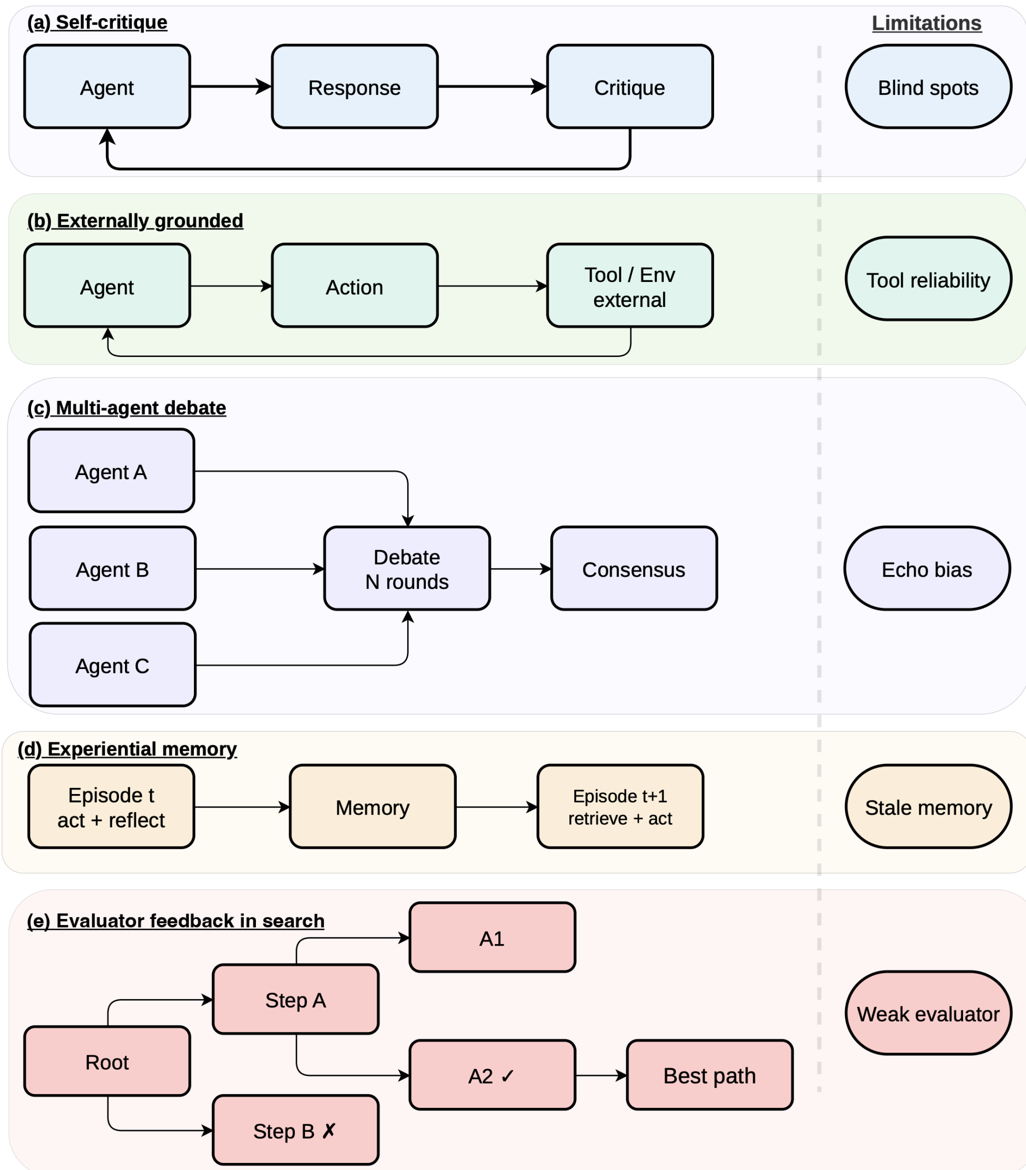

Fig. 2. Inference-time verbal feedback mechanisms: self-critique, externally grounded critique, multi-agent debate, experiential memory, and evaluator feedback in search.

models finds that debate often fails to beat chain-of-thought or self-consistency despite substantially higher inference cost, with model heterogeneity the most reliable intervention [332]. Shared biases produce redundant consensus [63, 146], and simpler single-agent strategies remain competitive [44].

Communication topology controls part of that cost. Full broadcast incurs substantial token overhead [44]; sparse connections reduce it [144]. Other methods retain only maximally disagreeing responses per round [184] or ask agents to express confidence so a knowledgeable minority does not capitulate to the group [152].

Debate also serves scalable oversight. It raises weaker model and human judge accuracy above naive baselines [123], and it can outperform consultancy under information asymmetry [121]. When transcripts become training data, rewards, or distilled weights, the same mechanism moves from deliberation to learning [182, 196, 233, 312].

2.2.3 Experiential-memory systems. Dedicated surveys catalog episodic, semantic, and procedural memory stores [58, 343]. The narrower VRL question is whether a stored linguistic artifact is consumed as feedback in a later context. Deployment-time learning has also been proposed as a lifecycle stage after pretraining and post-training [86].

Raw reflections are inexpensive and fit closely recurring tasks. Reflexion reports a HumanEval pass@1 of 91 percent through stored self-reflection [222]. When experience must transfer, distilled principles remove episode-specific detail. ExpeL contrasts successes and failures; MetaReflection compresses failed trials; related systems induce reusable workflows or strategy notes [87, 190, 239, 268, 347]. Structured stores become useful when scale or collaboration makes flat retrieval expensive. Interlinked notes, consolidation, team graphs, and reconstruction-time retrieval trade management overhead for selective access [43, 116, 295, 330]. Tasks requiring reliable re-execution favor code and instruction libraries whose skills can be tested and admitted with evidence [67, 148, 154, 251, 305, 344, 355].

Writing policy is a second design axis. Append-only stores suit short, stationary deployments. Longer-lived agents need utility-based pruning [23, 341]. Memory-R1, AutoMem, and SkillRL train add, update, delete, or no-op decisions [276, 284, 298]. Meta-evolution searches the entire encode, store, retrieve, and manage pipeline [155, 331, 333]. These writers can be trained by scalar outcomes while their stored text still acts through the context window.

Persistence amplifies both value and error. Large stores complicate retrieval [197]; stale knowledge misleads in changing environments [343]; and injected memories cause cross-session drift [54]. Forced consolidation can itself corrupt evidence, with raw-episode retention outperforming continual summarization in one controlled study [329]. Memory evaluation increasingly probes retrieval, test-time learning, long-range understanding, and selective forgetting [14, 101, 241, 272].

2.2.4 Evaluator feedback in search. Search systems explore several candidate paths and use language to select which to expand or prune [248]. Tree of Thoughts adds self-evaluation to a tree over intermediate thoughts [269, 308]; Reasoning via Planning and Graph of Thoughts generalize the topology and permit aggregation or refinement across branches [16, 90]. Node-level verbal evaluation can terminate bad paths early, but repeated evaluation multiplies model calls relative to single-pass generation [290]. This catalog excludes search procedures whose selection signal contains no language, including majority voting over sampled chains.

2.3 Extended Catalog: Language as Learning Signal

2.3.1 Feedback-conditioned and self-improvement systems. Feedback-conditioned modeling trains on triples containing an input, critique, and revised output. Early methods discarded the critique after using it to produce the target [28, 211]. ALT retains it as conditioning context [162], while the feedback-conditional policy learns the response posterior given feedback and adds an online positive-feedback bootstrapping stage [168]. Critique use can itself be trained: didactic multi-turn interactions teach a smaller model to predict and incorporate teacher feedback [128]. RLVF scopes a verbal instruction by generating preference data that mark where it should and should not apply [230]. The corresponding risk is over-receptiveness, since models can learn to follow incorrect critiques [216].

Self-improvement uses feedback to select generated data [225]. Filters include final-answer correctness in STaR [321], constitutional compliance [10], majority agreement [105], and a model's own iterated judgment [318]. Generator-judge coupling can amplify systematic bias over rounds [224]. Multiagent finetuning preserves diverse reasoning by training a society of models on interaction data [233]. Agent-R replaces whole-trajectory selection with targeted recovery, using tree search to connect failed paths to adjacent successful ones [317].

2.3.2 Trained critics. Critics obtain supervision from several sources. MultiCritique distills feedback from several agents into fine-tuning and preference data [135]. CTRL trains a critic to maximize the correction performance of a fixed code generator [291]. Critique-RL separates discriminability from helpfulness by first reinforcing rule-based error detection and then adding refinement reward [282]. DeepCritic combines long-form critique training with reinforcement over human or Monte Carlo step labels [303], while AutoMathCritique trains an actor-critic pair on automatically generated step feedback [283]. Critique-guided improvement jointly trains a critic and an actor so feedback steers exploration rather than only ranking final responses [302]. These systems show why feedback quality must be measured by both error localization and the correction it induces.

2.3.3 Multi-agent and self-play training. MALT samples role-specific search trees for a generator, verifier, and refiner, then propagates outcome rewards to each role through value iteration [182]. MAPoRL co-trains participants against correctness and collaboration incentives [196]. Self-play removes external annotation through adversarial word games [41], games a model plays against itself [131], or proposer, solver, and judge roles instantiated from one model [37]. SPC evolves a critic against a generator trained to produce difficult-to-detect reasoning errors [31].

The communication channel can itself receive reward. Influence-based rewards elicit accusation and evidence-giving in social deduction [210], while dialogue-game environments use interactive group-relative optimization [99]. Persuasion-balanced training prevents debaters from learning to reject beneficial persuasion along with adversarial persuasion [229]. Explicit debate can also be internalized: latent agents distill transcripts into a single model [312]. Related oversight systems reward prover checkability [127], use debate outcomes in sequential decision making [234], train debaters through self-play [7], or use debate for weak-to-strong supervision [137].

2.3.4 Process supervision and preference shaping. The canonical process-supervision dataset uses positive, negative, or neutral click labels for individual reasoning steps [147]. Such labels are scalar, but generative PRMs consume language before scoring [159, 348]. Step-level critique also qualifies when downstream machinery converts its content into credit. Process feedback improves localization relative to outcome-only labels [147, 245], and the cost of manual step annotation motivated automatic labeling with Math-Shepherd and OmegaPRM [167, 255]. Its practical limits remain expensive verification and difficulty defining a locally correct step outside mathematics and code.

Preference shaping is the maximally compressed endpoint. Preference-based reinforcement learning supplies the lineage [45]; InstructGPT trains a reward model and optimizes it with proximal policy optimization [189, 214, 231]; and DPO removes the explicit reward model [203]. KTO, ORPO, SimPO, and related objectives modify the implicit preference loss [8, 64, 97, 180], while RLAIF replaces human labels with model judgments [138]. Online optimization queries fresh policy samples, fixed offline datasets inherit coverage gaps, and iterated preference collection occupies the middle ground [318]. Open problems for this scalar endpoint are surveyed separately [24].

Across the compression spectrum, quantitative evidence is method-specific rather than directly comparable. GOLF reports improved sample efficiency by injecting group-level critiques into sparse-reward regions [107]; Text2Grad converts free-form critiques into span-level rewards [252].

These results motivate controlled comparisons that hold model, data, feedback budget, and verifier strength fixed.

2.4 Extended Catalog: The Verbal Reward Stack

2.4.1 Verifiable outcomes and rubric supply. Tulu 3 combines supervised fine-tuning, preference optimization, and reinforcement learning against programmatic checks in an open post-training pipeline [133]. DeepSeek-R1 reports emergent reflection and verification under outcome-only reinforcement learning [83]. Verifiable rewards remain sparse and domain-limited. Agent-RLVR adds teacher-style plans and error feedback to failed software-engineering attempts, then trains a reward model on the guided data [49]. Budget mismatches, attempt inflation, calibration drift, and benchmark contamination can confound headline RLVR gains [274].

Rubric rewards replace a ground-truth check with instance-specific criteria. Rubrics as Rewards reports gains over direct Likert-style judging in health and reasoning tasks [82]; checklist feedback improves hard instruction satisfaction [246]. Static banks support broad, stable task distributions, with human, synthetic, coarse-to-fine, or contrastively generated criteria [113, 143, 158]. Dynamic rubrics adapt to the current policy or case mix through online generation, policy-coupled evolution, self-rewarding, or case-conditioned criteria [81, 206, 256, 311]. Selective verification invokes external checks only for likely violations [275]; process-oriented rubrics detect invalid derivations that final-answer checks miss [319]. Prior work also surveys rubrics across evaluation, training, and safety [30].

2.4.2 Generative reward models. Generative verifiers cast reward modeling as next-token prediction and write a verification rationale before the verdict [336]. DeepSeek-GRM trains principle generation and critique, then allocates inference-time samples with a meta reward model [161]. Related systems generate rubrics or reference solutions before judging [34], spend adaptive compute on hard decisions [84], turn judgment into a verifiable-reward task [273], or separate evaluation planning from execution [208]. Other variants optimize judgment traces with offline or online reinforcement learning [103], extend judges to long-horizon reasoning [95], or pair a reasoning reward model with a dedicated benchmark [357].

Judge supervision is increasingly self-generated. Self-taught evaluators iterate without human preference labels [259]; joint critique and scalar prediction improves reward accuracy [316]; self-training produces reward models that emit labels with rationales [249]; and meta-rewarding asks the model to judge its own judgments [277]. The explanation is not automatically trustworthy. Outcome-only training can produce correct verdicts supported by unsound critiques, while explicit critique supervision improves reliability [267].

2.4.3 Process reward models. Human step annotation established the PRM lineage [147]. Math-Shepherd automates labels and applies stepwise policy optimization [255]; OmegaPRM uses divide-and-conquer search to locate the first error and build a large step-level dataset [167]. ProcessBench finds weak generalization beyond common mathematics benchmarks and cases where prompted critics outperform trained PRMs [350]. A companion analysis shows that Monte Carlo labels can underperform judge or human annotations, that best-of-N evaluation drifts toward outcome scoring, and that consensus filtering improves label quality [345].

Generative PRMs join step verification with language reasoning. ThinkPRM generates a verification chain for every step [122]; StepWiser trains stepwise reasoning with reinforcement from rollout outcomes [292]; and DG-PRM selects fine-grained criteria dynamically per step [314]. SPARK synthesizes verification data for domains without directly checkable answers [204], and related pipelines extend automatic step annotation to multimodal reasoning [57]. Response-level rubrics can also be converted to token-level credit through a relevance discriminator [294].

2.4.4 Boundary failures and defenses. The scalarization boundary creates exploitable slack at every rung. Outcome-only optimization can encourage invalid "Miracle Steps" [319]; static rubrics become stale or omit failure modes [206]; and rubric-based verifiers can prefer a trained checkpoint even when rubric-free judges prefer the base model [176]. Generative judges can reach correct verdicts from unsound critiques [267], and misestimated step labels corrupt PRM training [345]. RLVR results can also shrink under budget-matched evaluation [274].

Proposed defenses operate at the same boundary: cross-family judge panels and verifier-free diagnostics [176], peer consensus over evolving rubrics [81], format constraints on PRM-driven reinforcement learning [204], and consensus filtering of step labels [345]. These methods reduce known failures but do not remove the information lost when critique text becomes a scalar.

3 Extended Evidence on Credit Assignment

This section records implementation details and paper-specific empirical evidence that support the credit-assignment synthesis in the main article. Reported numbers are not directly comparable because the methods use different benchmarks and baselines.

3.1 Token- and Step-Level Mechanisms

MA-RLHF performs policy-gradient updates over macro actions, sequences of tokens or higher-level constructs, shrinking the temporal distance between action and reward without extra inference compute. Its authors report gains of up to 30% on summarization and code generation, 18% on dialogue, and 8% on question answering, reaching parity with vanilla RLHF 1.7–2 times faster in training time [25]. RED instead reallocates a sequence-level reward to individual tokens with an existing reward model and no additional training stage [141]. LaRe generalizes redistribution to episodic settings through a latent, multidimensional performance evaluation constructed from LLM-generated executable code [202].

Model-internal methods derive credit from the scorer itself. One approach estimates per-token rewards from SHAP and LIME attributions and uses Bayesian optimization to tune the shaping function; feature-additive attributions preserve the optimal policy of the original reward [129]. FlowTracer builds an attention-induced graph over tokens and uses information flow toward the answer as a shaping signal [55].

Critique-derived methods make localization explicit. CAPO generates all stepwise critiques in one pass, converts them into deterministic token credits, and votes over critiques for robustness [288]. T-REG uses contrastive prompting to self-generate token rewards that regularize the distribution of sequence-level preference reward [359]. iStar alternates policy optimization with an implicit process reward model and derives step rewards without extra rollouts or step labels [159]. Text2Grad aligns free-form critiques with the token spans they indict and backpropagates span-level rewards [252].

3.2 Multi-Turn and Memory-Conditioned Credit

ArCHer combines an off-policy value learner across utterances with a within-turn token-level policy-gradient learner; its authors report roughly 100 times the sample efficiency of prior methods on their agent tasks [364]. Multi-turn variants of PPO and group-relative optimization show that well-designed turn rewards can improve stability and convergence over trajectory-only rewards [271]. IGPO uses the marginal increase in the policy's probability of the correct answer as an intrinsic turn reward, requiring no reward model or Monte Carlo estimate [250]. LMRL-Gym provides multi-turn dialogue and text-game tasks for studying this regime [1].

Memory turns write operations into actions that also require credit. Memento realizes policy improvement through memory rewriting and retrieval [354]. Memory-R2 shows that writes cause

Table 3. Numerical optimization concepts and their language-space counterparts.

Numerical concept	Language-space counterpart	Representative realizations
Parameters	Prompt, evolving context, or curated playbook	VML [286]; ACE [338]; token priors [22]
Gradient	Natural-language critique of the current candidate	TextGrad [320]; directional feedback [185]
Update rule	LLM edit operator applied to text	OPRO [300]; GEPA [2]
Backpropagation	Feedback routed through a computation graph	Trace/OPTO [39]; semantic backpropagation [261]; LLM-AutoDiff [313]
Momentum or curvature	History of responses and scores	REVOLVE [337]; OPRO [300]
Ensemble	Population of candidates under selection	PromptBreeder [72]; EvoPrompt [85]; GEPA [2]

rollouts to diverge in their effective environment, violating assumptions behind group-relative baselines; it combines a global long-horizon objective with local rerollouts from the same intermediate memory state [297].

3.3 External Language Models as Credit Annotators

CALM decomposes a task into subgoals and judges whether transitions achieve them, creating auxiliary option-termination rewards without examples or fine-tuning [200]. Language feedback models learn from LLM judgments over verbalized trajectories and identify actions for imitation learning, including in unseen environments [351]. In multi-agent RL, LLM-generated dense, agent-specific rewards can be stabilized through potential-based reward learning over multiple queries [149]. Group-level critique can also guide exploration through off-policy scaffolds rather than assign credit directly [107]. These methods are most useful when the policy is not itself a language model or when an existing RL pipeline must remain unchanged, but annotator errors then become systematic reward errors.

4 Extended Systems, Evaluation, and Safety Catalogs

This supplement preserves the method catalogs, benchmark matrix, quantitative evidence, and robustness studies condensed in the main article. Its organization follows the four main-paper topics but uses independent labels so it can be compiled as part of the electronic supplement without relying on main-paper cross-references.

4.1 Language-Space Optimizer Catalog

Language-space methods instantiate a recognizable optimization loop in text. Table 3 records the correspondence used to classify them.

4.1.1 Search, Evolution, and Program Optimization. APE uses an LLM to propose instructions and selects candidates by task performance, establishing the basic score-driven recipe [361]. OPRO adds the optimization trajectory itself to the optimizer context: previous solutions and their scores become working memory, and the resulting prompts outperform human-designed prompts by up to 8% on GSM8K and 50% on Big-Bench Hard [300]. Evolutionary methods maintain populations instead of one trajectory. EvoPrompt treats LLMs as mutation and crossover operators over a prompt population [85]; PromptBreeder also evolves the mutation prompts that generate task-prompt variants [72]. At the program level, DSPy compiles declarative language-model modules toward a metric by generating and selecting demonstrations [124]. MIPROv2 jointly optimizes instructions and demonstrations across multi-stage programs without module-level labels, so its surrogate model must attribute program-level success to components [188].

4.1.2 Textual Gradients and Trace-Level Credit. TextGrad attaches natural-language critiques to variables in a compound computation and routes them backward as textual gradients; without task-specific modification it raises GPT-4o zero-shot GPQA accuracy from 51% to 55% [320]. Meta-TextGrad extends this recursion to the optimizer's own prompts and structure [293]. Directional feedback stabilizes updates by stating how the candidate should move, not merely why it is wrong [185]. REVOLVE retains the evolution of responses across iterations rather than reacting only to the latest critique [337]. Semantic backpropagation allocates output feedback among graph nodes that share a successor [261], and LLM-AutoDiff extends routing to cyclic, multi-component workflows [313]. Textual gradients can still explode as feedback grows with depth or vanish when repeated compression removes specificity; textual equilibrium propagation replaces one global backward pass with local critic equilibria [32].

The same machinery can optimize system structure. Agent symbolic learning jointly updates prompts, tools, and pipeline structure through language losses and gradients [358]. ADAS searches over agents expressed as code [100]. AFlow casts workflow generation as Monte Carlo tree search over code-represented workflows and reports smaller models outperforming GPT-4o on selected tasks at 4.55% of its inference cost [334]. MASS interleaves prompt and topology optimization for multi-agent systems [356].

4.1.3 Reflective, Training-Free, and Hybrid Updates. GEPA samples system trajectories, diagnoses failures in language, proposes prompt updates, and combines complementary lessons from a Pareto frontier. It reports a 6% average and up to 20% advantage over GRPO while using up to 35× fewer rollouts, plus a greater than 10% gain over MIPROv2 [2]. Self-supervised prompt optimization derives evaluation and optimization signals from pairwise output comparisons, matching leading optimizers at 1.1%–5.6% of their cost with as few as three samples [285]. Training-Free GRPO extracts a group-relative semantic advantage from rollouts and distills it into a token prior for a frozen model [22]. ACE maintains an evolving playbook through generation, reflection, and curation; incremental updates target brevity bias and context collapse in wholesale rewriting [338].

Hybrid systems decide where knowledge should live. SEAL generates finetuning data and update directives in language before changing weights [365]. Learning to Self-Evolve trains the context editor with reinforcement learning and reports a 4B editor outperforming larger self-evolution systems [36]. P^2O alternates policy-gradient updates with prompt evolution, using text search when all sampled advantages collapse and then distilling prompt-induced gains into weights; it reports up to 9.5% improvement over GRPO baselines [165]. Evolutionary system-prompt learning finds declarative knowledge accumulating in prompts while procedural knowledge moves into weights [335]. SIA extends joint updating to scaffold code, prompts, and weights [93].

4.1.4 Failure Evidence and Boundary Cases. Reflective optimizers can substitute prior knowledge for causal error analysis, producing confident but misdirected edits [172]. Seed quality matters: from a defective seed, GEPA reduced GSM8K accuracy from 23.81% to 13.50%, whereas VISTA separated hypothesis generation from rewriting and verified hypotheses on minibatches, recovering 87.57% [157]. Multi-objective critics suffer gradient dilution and instruction interference; one study reports a 59% drop in task focus when a single critique addresses several criteria [51]. MASPromptBench further finds that gains vary substantially across workflows, protocols, and team sizes [9]. GReaTer uses actual numeric task-loss gradients to optimize prompts [52], while PromptQuine evolves text through a selection signal with no linguistic meaning [254]. Both optimize text but fall outside verbal reinforcement learning because their improvement signal is non-verbal.

4.2 Extended Embodied and Multimodal Evidence

4.2.1 Corrections on Real Robots. LILAC maps utterances onto a learned low-dimensional control space used for real-time shared autonomy on a Franka Emika Panda; users preferred it and achieved higher completion than with open-loop instruction following [48]. DROC accepts corrections ranging from high-level planning failures to skill parameters, distills them into a knowledge base, and retrieves them by textual and visual similarity; it required roughly half the corrections of direct code-generation baselines in the first round and little to none after two iterations [324]. Yell At Your Robot logs fine-grained corrections and folds them into iterative retraining, improving long-horizon dexterous manipulation without additional teleoperation [221]. RT-H inserts language motions between task and action, allowing people to intervene in the same vocabulary used by the hierarchy; language interventions outperformed teleoperated ones [15]. RACER replaces the person with a vision-language supervisor that produces recovery guidance and improves over an imitation baseline in RLBench and real-world trials [50]. Inner Monologue verbalizes success detection, scene description, and human feedback to improve instruction completion without training [109].

4.2.2 Generated and Vision-Language Rewards. Language to Rewards compiles instructions and corrections into reward parameters for a real-time optimizer [315]. Text2Reward generates dense reward code that can be refined through human feedback and transfers simulator-trained policies to the real world [289]. Eureka mutates reward programs from verbalized training statistics [173]; DrEureka adds domain-randomization generation and reports sim-to-real behaviors competitive with hand-designed setups [174].

When only pixels are available, RL-VLM-F learns rewards from vision-language preference labels over image observations [266]. VLM-RL converts contrasting language goals into dense semantic rewards for driving, reducing collision rate by 10.5% and increasing route completion by 104.6% in CARLA [112]. VLA-RL trains a robotic process reward model over automatically segmented tasks, lifting OpenVLA-7B by 4.5% over its strongest fine-tuned baseline on LIBERO [164]. RoboReward augments success-heavy robot corpora with calibrated failures and finds that no existing VLM dominates its reward-assignment tasks [139]. Robo-Dopamine combines a step-aware process reward with policy-invariant shaping; after one-shot adaptation, it improves a near-zero policy to 95% success within 150 online rollouts [240]. Large Reward Models emit process, completion, and temporal-contrastive rewards in unseen environments [279], while SOLE-R1 reasons over raw video and language goals at each step and reports zero-shot online learning on 24 unseen tasks [213].

Two benchmarks isolate perceptual limits. ViSTa finds that models recognize objects but usually fail to understand task sequence [280]; MotIF shows that final-frame detectors fail when success depends on motion and improves them through trajectory-overlaid representations [114].

4.2.3 Failure Explanations and Recovery. AHA fine-tunes a vision-language model on perturbed demonstrations to explain failures in free-form language. It reports a 10.3% advantage over GPT-4o in-context learning and a 21.4% average downstream improvement when explanations guide reward refinement, planning, and verification [61]. FailSafe pairs generated failures with executable recovery actions and improves three vision-language-action models by up to 22.6% on average [150]. Guardian synthesizes failures from simulated and real trajectories with structured reasoning traces and raises manipulation success when inserted into a policy loop [191]. Controlled experiments also show that diverse and informative language improves generalization and adaptation, making feedback phrasing an experimental variable rather than incidental wording [281].

Table 4. Benchmark families indexed by quality or utilization, signal type, stage, and principal gap.

Benchmark family	Measures	Signal	Principal gap
Reward-model benchmarks: RewardBench 1/2 [134, 177], RM-Bench [160], RewardMATH [125], RMB [353], PPE [74]	Quality	Outcome preference	Pairwise accuracy can decorrelate from downstream utility
Judge meta-evaluation: LLMBar [323], JudgeBench [242], CALM [310], MM-Eval [227], Sage [70], BiasScope [132]	Quality	Verdict and rationale	Bias, inconsistency, and aggregation uncertainty
Critique benchmarks: CriticBench [151], CriticEval [136], CodeCriticBench [327], RealCritic [243]	Both	Natural-language critique	Open-loop ratings can diverge from corrective value
Process and verifier benchmarks: ProcessBench [350], PRMBench [228], ToolComp [183], VerifyBench [299]	Quality	Step-level reward or verdict	Narrow domains, false negatives, and hackability
Interactive environments: MINT [265], ConvCodeWorld [89], LLF-Bench [38]	Utilization	Turn-level feedback	No standard metric for uptake from language alone
Feedback incorporation: Feedback Friction [117], JETTS [363]	Utilization	Near-oracle or judge critique	Models under-absorb accurate feedback
Agent trajectory and memory: AgentRewardBench [170], Plan-RewardBench [253], MemoryAgentBench [101]	Both	Trajectory preference and memory	Weak long-horizon judgment and retrieval
Embodied feedback: ViSTa [280], MotIF-1K [114], RoboReward [139]	Quality	Vision-language reward	No embodied utilization benchmark

4.3 Extended Evaluation Matrix and Evidence

4.3.1 Quality Evidence. RewardBench establishes a broad outcome-preference test across chat, reasoning, and safety [134]. RewardBench 2 uses new human prompts and reports average model scores about 20 points lower while retaining correlation with best-of-N and PPO outcomes [177]. RM-Bench separates content sensitivity from style susceptibility and finds state-of-the-art reward models averaging 46.6% under style interference [160]. RewardMATH shows that single chosen-rejected comparisons can have almost no correlation with optimized-policy results, while one-to-many comparisons recover useful correlation [125]. RMB expands to more than 49 scenarios [353]; Preference Proxy Evaluations validate metrics against end-to-end RLHF [74]; Personalized RewardBench tests user-specific rubrics [171].

LLMBar constructs pairs where an incorrect answer has deceptive surface appeal [323], and JudgeBench finds GPT-4o-level judges only slightly above random on correctness-grounded hard pairs [242]. CALM measures twelve judge biases through controlled perturbations [310]. BiasScope discovers biases automatically and reports error above 50% even for strong evaluators on its hardened set [132]. Sage finds inconsistent frontier-model preferences in nearly a quarter of difficult cases [70]; leaderboard audits show Elo aggregation masking ranking uncertainty [73]. MM-Eval documents weaker non-English judging across 18 languages [227].

CriticBench evaluates generation, critique, and correction together [151]; CriticEval adds human reference critiques [136]; CodeCriticBench uses checklist-scored code critique [327]. ProcessBench asks for the earliest erroneous reasoning step [350], PRMBench probes finer error dimensions [228], and ToolComp tests multi-tool trajectories where most models score below 50% [183]. VerifyBench covers reference-based RLVR checkers [299]; rule-based verifiers can reject equivalent answer formats, while model-based verifiers are more accurate but hackable [111].

4.3.2 Utilization Evidence and Coverage Gaps. RealCritic grades critiques by induced corrections and finds models falling below their own baselines in some self-critique settings [243]. JETTS finds judges competitive with outcome reward models for reranking but worse than process reward models in beam search; their textual critiques do not reliably guide generators [363]. MINT reports tools adding 1–8% absolute performance per turn and simulated language feedback adding 2–17% [265]. ConvCodeWorld varies compilation, execution, and verbal feedback across nine code-repair scenarios, with static replay preserving Spearman correlations of 0.82–0.99 with live interaction [89]. LLF-Bench isolates learning from language across attempts [38], while LMRL Gym provides a numeric-reward complement over text states and actions [1]. Feedback Friction shows that near-complete ground-truth feedback can still fail to change solver behavior [117].

AgentRewardBench compares twelve judges with expert annotations of 1,302 web trajectories and finds no judge best across benchmarks [170]. Plan-RewardBench documents weakness across generative rewards, discriminative rewards, and judges on tool-integrated plans [253]. MemoryAgentBench tests retrieval, test-time use, and selective forgetting [101]. Coverage remains thin for non-English feedback [227], personalization [171], long-horizon agents [253], memory [101], and embodied utilization [114, 139, 280].

4.4 Extended Threat and Defense Evidence

4.4.1 Judge Manipulation. Optimized injected sequences can force a judge to select an attacker-controlled candidate across evaluation, search, tool selection, and AI-feedback training [219]. Universal phrases transfer from surrogate to unseen judges and push absolute scores toward their maximum, with comparative assessment less susceptible [205]. Generative reward models accept superficial “master keys” that elicit false positive rewards across scales [349]. Prompt-injection studies report attack success up to 73.8% and cross-judge transfer of 50.5%–62.6% [178]; robustness varies by up to 40% across prompt templates [142]. Stylistic manipulation exceeds 65% success while preserving semantics [304]. Automated bias discovery pushes strong evaluators above 50% error [132]; apologetic phrasing skews some safety evaluations by up to 98% [29]; and red-team distribution shifts can reduce judge accuracy toward chance [215].

4.4.2 Observation Injection and Feedback Poisoning. Indirect prompt injection embeds instructions in retrieved content [80]. InjecAgent reports 24% attack success against a ReAct-prompted GPT-4 agent, nearly doubling when instructions are reinforced [326]. AgentDojo separates task failure from security failure across tool-integrated scenarios [53]. Human-written web injections partially succeed in up to 86% of cases even when agents fail to complete the attacker’s full objective [65]. A public competition spanning 272,000 attempts against 13 frontier models found every model vulnerable across tool use, coding, and computer control, with 0.5%–8.5% end-to-end success and transferable strategies [62]. Injection can steer tool choice [220], and adaptive attacks bypass evaluated defenses [325].

Training-time poisoning works at low rates. Adding 1–5% poisoned preference pairs can steer target sentiment [12]; DPO backdoors succeed with 0.5% poisoning in settings where PPO requires at least 4% [198]. A model can also influence preferences constructed from its own outputs, amplifying keyword and brand biases [88]. Manipulated verbal feedback raises language-space optimizer attack success by as much as 0.48, and fake-reward attacks need no reward-model access [346].

4.4.3 Policy Exploitation and Sycophancy. When model outputs change the world and the world returns to context, output and policy refinement can produce in-context reward hacking that static evaluation misses [192]. Models trained to game harmless graders can transfer hacking to new settings and, for some models, to shutdown evasion [244]. Rubric exploitation clusters around partial satisfaction, implicit content treated as explicit, and imprecise topical matching [176]. Controlled tasks with injected judge biases [263] and verifiable hacking opportunities [207] make the behavior measurable. Adversarial preambles can raise frozen candidates' scores, transfer to unseen judges, and remain difficult to detect [5].

Sycophancy originates in preference data rather than an external attacker. Humans and preference models sometimes prefer convincing agreement to correctness [216]. Across mathematics and medical advice, one study reports sycophantic answer changes in 58.19% of cases and regressive flips in 14.66% [68]. Multi-turn pressure compounds the behavior, while a third-person perspective reduces it by up to 63.8% in debate [96]. Models preserve a user's desired self-image more than humans and affirm both sides of moral conflicts [40]. Mechanistic analysis identifies a circuit that represents an answer as wrong even while the model agrees [194].

4.4.4 Defense Evidence. Perplexity filters and known-answer checks fail against optimized injections, although re-tokenization and LLM detectors are stronger among tested detection options [142, 219]. Signal reshaping acts earlier: verifier training with truncated adversarial negatives resists master keys [349]; adversarial reward-model training searches for high-reward, low-quality responses and stabilizes downstream RLHF [21]; linear-probe penalties suppress sycophancy markers [195]; and uncertainty-aware trajectory rewards reduce sycophancy without degrading general capability [13].

System defenses include instruction hierarchies [247], preference optimization against injection [33], and constrained agent-tool design patterns [17]. Highlighting untrusted optimizer feedback cuts one fake-reward attack's success delta from 0.23 to 0.07 without reducing utility [346]. No single layer is sufficient because adaptive attacks have bypassed targeted defenses [325]. The evidence favors defense in depth: comparative scoring, independent judges, provenance marking, privilege separation, adversarially reshaped rewards, and interactive stress tests.

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